

Requirements Engineering & UML Use-Case Modelling

PES1UG24CS144-Clinical Trial Subject Recruitment Engine

Actors: Trial Candidate, Clinical Researcher, Notification Service.

ID	Type	Description	Priority	Acceptance Criteria	Rationale
FR-001	Functional	The system shall compare a trial candidate's questionnaire answers against the inclusion and exclusion criteria of the selected trial and mark the candidate as Eligible or Not Eligible.	High	Pass: A candidate who meets an exclusion criterion is marked Not Eligible and the failed criteria are listed. Fail: Such a candidate is marked Eligible.	Automated pre-screening is the main purpose of the system and prevents unsafe enrolment.
FR-002	Functional	The system shall allow a Clinical Researcher to add and edit the inclusion and exclusion criteria of a trial.	High	Pass: Criteria saved by the researcher are used in the very next screening. Fail: Screening still uses the old criteria.	Trial criteria are revised during recruitment and must stay up to date.
FR-003	Functional	The system shall display the informed consent form to an eligible candidate and store the candidate's electronic signature along with the form version, date and time.	High	Pass: The consent record contains the signature, form version and timestamp, and enrolment is blocked until it exists. Fail: A candidate is enroled without a stored consent record.	Informed consent is legally required before a subject can join a trial.
FR-004	Functional	The system shall allow a consented candidate to book study visits and allow the Clinical Researcher to mark each visit as Attended or Missed.	Medium	Pass: A visit the subject did not attend appears as Missed in the visit compliance log. Fail: The visit is shown as Attended or is not recorded at all.	Visit compliance has to be tracked to detect protocol deviations.
FR-005	Functional	The system shall show the Clinical Resercher a report giving the number of candidates screened, found eligible, consented and enrolled for a selected trial.	Medium	Pass: Every count in the report matches the stored candidate records. Fail: Any count differs from the stored records.	Researchers need to track recruitment progress against the enrolment target.
NFR-001	Non-functional (Security)	The system shall record an audit trail entry containing the user, date, time, old value and new value for every change made to a candidate, eligibility or consent record, as required by 21 CFR Part 11.	High	Pass: Every record change creates an audit entry, and stored entries cannot be edited or deleted. Fail: Any record change leaves no audit entry.	Regulators require proof that clinical records have not been altered.
NFR-002	Non-functional (Performance)	The system shall display the eligibility result within 3 seconds for at least 95% of submissions while 200 candidates are using the system at the same time.	High	Pass: A load test with 200 concurent users returns 95% of results in 3 seconds or less. Fail: More than 5% of results take longer than 3 seconds.	Slow screening feedback causes candidates to abandon the application.